

Annotation Matters: Evaluating Grad-CAM and Bounding Box Weak Supervision under Modular Segmentation with CRF Refinement

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1 Introduction

Semantic segmentation is a fundamental task in computer vision, aiming to assign a class label to every pixel in an image. While fully-supervised methods have achieved impressive performance, they heavily rely on dense pixel-level annotations, which are expensive and time-consuming to obtain.

To reduce annotation costs, research has shifted towards *weakly-supervised semantic segmentation (WSSS)*, where models are trained using coarse labels such as image-level tags [4], bounding boxes [3], or saliency maps. Among these, **Grad-CAM** [4] generates class-specific activation maps from classifiers and has been used to provide foreground localization. Alternatively, **bounding boxes** [3] offer slightly more spatial context while remaining cost-efficient.

Our project builds upon the SEC framework [2], which introduced a composite loss—*seeding* (encouraging correct class predictions in confident regions), *expansion* (promoting class coverage beyond seeds), and *boundary constraint* (refining mask alignment with object edges)—to train segmentation networks using CAM-derived masks. However, instead of SEC’s joint optimization, we adopt a **modular pipeline**: we first train a classifier to generate Grad-CAM masks, then independently train segmentation networks using these pseudo labels. This separation improves transparency, facilitates debugging, and enables structured experimentation across annotation types.

This study addresses two core questions: (1) **MRP**: Can Grad-CAM—when combined with modular decoupling and CRF post-processing—serve as an effective weak supervisory signal that narrows the performance gap with fully-supervised segmentation? (2) **OEQ**: What performance trade-offs arise when replacing Grad-CAM masks with bounding-box-derived labels, and how does this substitution reflect on segmentation quality under the same training pipeline?

By benchmarking these setups against a fully-supervised baseline, we aim to quantify how annotation type and supervision granularity influence model performance in a practical, modular segmentation framework.

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2 Methods

To investigate how different weak annotation forms affect segmentation performance under a modular architecture, we implemented a two-stage framework inspired by SEC [2], but with a full decoupling of the classification and segmentation phases. Our pipeline includes three key modules: (1) classification for pseudo mask generation, (2) segmentation using weak labels, and (3) optional CRF refinement.

2.1 Classification and Grad-CAM-based Mask Generation

We adopt VGG16 as the backbone classifier to predict image-level labels on the Oxford-IIIT Pet dataset. Following [4], we use **Grad-CAM** to extract class-specific activation heatmaps from the final convolutional layer. These Grad-CAMs are then thresholded (default: 0.2) to generate foreground localization masks, serving as weak pixel-level annotations.

Unlike SEC [2], which integrates CAM extraction into a joint training process, we train the classification model independently, freeze its weights, and extract Grad-CAMs post-hoc. This architectural decoupling provides clearer diagnostic access to classification performance and pseudo-label quality, and facilitates replacing Grad-CAMs with other weak label sources (e.g., bounding boxes).

2.2 Weakly-Supervised Segmentation Network

The segmentation module adopts DeepLabV3+ with a VGG16 backbone, trained solely on the pseudo masks produced in the first stage. We initially experimented with three backbone architectures—FCN [1], UNet [1], and DeepLabV3+ [5]—and found that DeepLabV3+ consistently achieved the highest segmentation accuracy; hence, we adopted it as our primary segmentation framework. The network is optimized using a composite loss composed of:

- **Seeding Loss**: Encourages correct classification in confident Grad-CAM regions;
- **Expansion Loss**: Promotes mask coverage beyond the initial seed using class priors;
- **Constrain-to-Boundary Loss**: Applies CRF post-processing to refine boundaries, encouraging alignment with object edges.

Training is conducted under weak supervision only—no access to ground truth masks is permitted. The CRF module can be toggled to support ablation studies.

2.3 Bounding Box Supervision (OEQ Extension)

To address our open-ended research question, we substitute Grad-CAM-derived masks with bounding box-based masks [3] and re-run the segmentation training.

Each bounding box is converted into a rectangular region assumed to cover the target object, forming alternative weak labels. The training procedure, loss functions, and network configuration remain consistent, allowing direct comparison with the Grad-CAM-based setup.

This modularity enables fair benchmarking of different annotation strategies under identical segmentation conditions, directly testing the influence of annotation granularity.

3 Experiments

We evaluated multiple supervision strategies under a modular segmentation framework using the Oxford-IIIT Pet dataset. Segmentation performance was assessed using mean Intersection over Union (mIoU) and pixel accuracy. All models were trained under identical conditions for fair comparison.

3.1 Baseline Comparison: Fully vs. Weak Supervision

To establish an upper-bound benchmark, we trained fully-supervised models with three backbone architectures: FCN, UNet, and DeepLabV3+. DeepLabV3+ achieved the best mIoU (0.9227), followed by FCN (0.8320) and UNet (0.6840). Based on this, DeepLabV3+ was adopted as the default segmentation network.

We then trained weakly-supervised models using Grad-CAM-derived masks within our modular pipeline (Section 2), and compared them with models trained using bounding-box-derived labels [3].

As expected, fully-supervised models outperformed all weak variants. Nonetheless, both Grad-CAM and box-based supervision produced reasonable segmentation, demonstrating their potential in annotation-constrained scenarios.

3.2 Ablation Study I: CRF Post-Processing

To assess the contribution of CRF refinement [2], we trained the CAM-based segmentation model with and without CRF post-processing. All other settings were kept identical. As expected, the CRF-refined masks yielded better boundary precision and slightly improved mIoU. The visual results also confirmed sharper object contours in CRF-enabled predictions.

This study underscores the importance of post-processing in mitigating the noisiness of weak localization cues, especially in high-frequency regions where Grad-CAM tends to blur or diffuse boundaries.

3.3 Ablation Study II: CAM Foreground Threshold

Grad-CAM heatmaps require thresholding to binarize foreground regions. We conducted sensitivity analysis on the threshold value used to generate pseudo masks, comparing models trained with thresholds of 0.05, 0.1, and 0.2.

Empirically, we found that a low threshold (0.05) led to overly inclusive masks, sometimes extending into background regions. Conversely, a high threshold (0.2) produced sparse and conservative masks, omitting parts of the target. The intermediate threshold (0.1) yielded the best trade-off between coverage and precision, balancing segmentation accuracy and stability.

This ablation emphasizes the need to tune localization thresholding as an essential hyperparameter in CAM-based pipelines [4].

4 Results

4.1 Baseline Backbone Comparison: FCN vs. UNet vs. DeepLabV3+

We evaluated three backbones—FCN, UNet, and DeepLabV3+—under fully-supervised settings. As shown in Table 1, DeepLabV3+ outperformed the other models on all metrics, and was thus selected as the backbone for subsequent weak supervision experiments.

Table 1. Fully-supervised performance across different backbones.

Method	Mean BG IoU	Mean FG IoU	Mean IoU
FCN	0.7791	0.8848	0.8320
UNet	0.6001	0.7678	0.6840
DeepLabV3+	0.9023	0.9431	0.9227

4.2 Comparison Across Supervision Strategies

Table 2 summarizes the performance of weakly-supervised pipelines using Grad-CAM (with/without CRF) and bounding box masks. Grad-CAM with CRF yields the best weakly-supervised performance, while the naive Grad-CAM baseline performs the worst. Pixel accuracy is omitted for CAM-based methods due to foreground sparsity.

Table 2. Comparison of segmentation performance across supervision strategies.

Method	Mean IoU	Pixel Accuracy
Fully-supervised (DeepLabV3+)	0.9227	0.9686
Grad-CAM + CRF	0.7559	—
Grad-CAM (no CRF)	0.4767	—
Bounding Box	0.5043	0.7775

Note: Pixel accuracy is not applicable to Grad-CAM-based methods due to imprecise pixel-level labels.

4.3 CRF Post-Processing Effects

CRF refinement substantially improved Grad-CAM-based segmentation (Table 3), especially for boundary alignment. Without CRF, foreground masks become noisy and fragmented.

Table 3. Ablation on CRF post-processing in Grad-CAM pipeline.

Setting	FG IoU	BG IoU	Mean IoU
With CRF	0.6865	0.8254	0.7559
Without CRF	0.4834	0.4700	0.4767

4.4 Grad-CAM Threshold Sensitivity

Thresholding strongly affects Grad-CAM mask quality. Table 4 shows that 0.1 achieves the best trade-off between inclusion and precision, while extreme values (0.01 or 0.2) lead to under/over-segmentation.

Table 4. Grad-CAM foreground threshold ablation.

Threshold	FG IoU	BG IoU	Mean IoU
0.01	0.3814	0.1663	0.2739
0.05	0.6865	0.8254	0.7559
0.1	0.6202	0.8007	0.7105
0.2	0.5189	0.7685	0.6437

4.5 Visual Comparison (See Appendix)

5 Discussion

5.1 Key Findings

1. DeepLabV3+ Is the Best Supervised Backbone. Among fully-supervised architectures, DeepLabV3+ significantly outperformed FCN and UNet, justifying its selection as the default backbone.

2. Grad-CAM + CRF Performs Best Among Weak Labels. While weak supervision lags behind full supervision, Grad-CAM with CRF post-processing delivers the highest mIoU among weakly-supervised settings, showing the importance of spatial refinement.

3. Bounding Box Beats Raw Grad-CAM. Replacing Grad-CAM with bounding-box masks improves segmentation quality, confirming that geometric structure—even if coarse—is more effective than raw activation maps.

4. Thresholding Strongly Influences CAM Quality. Grad-CAM performance is sensitive to foreground thresholding; 0.1–0.05 achieves a favorable balance, while extreme values degrade accuracy.

5. CRF Sharpens Boundaries. Visual and quantitative results both show that CRF dramatically improves object boundary localization in Grad-CAM masks, reducing false positives and fragmentation.

5.2 Unanswered Questions

While bounding-box supervision outperforms raw Grad-CAM, it often exhibits boundary leakage and poor alignment around object edges. Moreover, both weak supervision types fail to capture instance-level separation, limiting effectiveness in scenes with overlapping or adjacent objects. It remains unclear how these methods would scale to multi-instance or real-world cluttered datasets.

5.3 Limitations

The Grad-CAM and bounding-box labels used are approximate by design, introducing noise that constrains peak model performance. Our study is conducted exclusively on the Oxford-IIIT Pets dataset, which features centered subjects and minimal occlusion. As such, the generalizability of these findings to more complex or large-scale benchmarks remains to be validated.

5.4 Future Work

Future efforts could refine bounding box masks using contour-aware post-processing or edge-preserving methods like CRF. Another promising direction involves self-training with pseudo-labels to iteratively denoise weak masks. Finally, extending the evaluation to urban-scale datasets such as Cityscapes or COCO-Stuff would provide stronger evidence of real-world applicability.

6 Conclusion

This study examined how annotation type and granularity shape the effectiveness of weakly-supervised semantic segmentation under a modular training framework. We showed that Grad-CAM alone produces noisy pseudo masks, but when paired with CRF refinement, it can substantially close the gap to fully-supervised performance—answering our main research question (MRP). Furthermore, substituting Grad-CAM with bounding-box-derived masks yields more spatially coherent segmentation and offers a viable alternative, albeit still inferior to full supervision (OEQ). These findings highlight that the success of weak supervision depends not only on label availability, but also on how structural priors and post-processing are incorporated. In this context, annotation indeed matters.

References

1. Kojec, J.: The oxford-iiit pets segmentation using unet. <https://github.com/Kojec1/The-Oxford-IIIT-Pets-Segmentation> (2023), accessed: 2025-04-23
2. Kolesnikov, A., Lampert, C.H.: Seed, expand and constrain: Three principles for weakly-supervised image segmentation. In: European Conference on Computer Vision (ECCV). pp. 695–711 (2016)
3. Lee, J., Kim, S.J.: Bbam: Bounding box attribution map for weakly supervised semantic and instance segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2643–2652 (2021)
4. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-cam: Visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV). pp. 618–626 (2017)
5. Wang, F.: Deeplabv3plus in pytorch. <https://github.com/VainF/DeepLabV3Plus-Pytorch> (2021), accessed: 2025-04-23

Appendix. Segmentation Visualizations

A1. Fully-Supervised Baseline Predictions

For the fully-supervised baseline, we experimented with three popular segmentation architectures: FCN, UNet, and DeepLabV3+. All models were trained using ground truth masks on the Oxford-IIIT Pet dataset under identical settings.

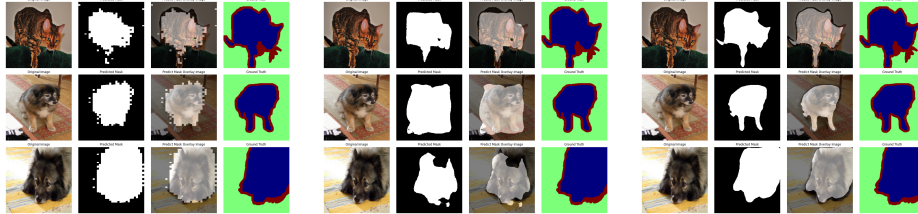


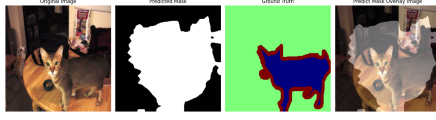
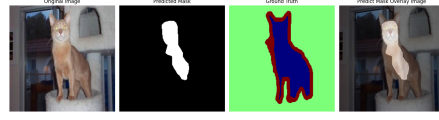
Fig. 1. *
Fig. A1.1 FCN
prediction results.

Fig. 2. *
Fig. A1.2 UNet
prediction results.

Fig. 3. *
Fig. A1.3 DeepLab
prediction results.

Observation: Among the three supervised baselines, DeepLabV3+ yields the most precise and visually complete segmentation results, particularly around object boundaries, indicating its superior feature extraction and refinement capabilities.

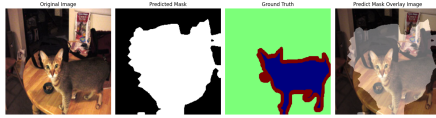
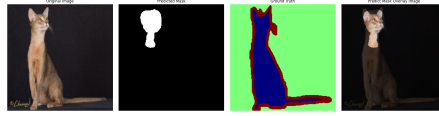
A2. Grad-CAM-Based Weak Supervision (with vs. without CRF)

**Fig. 4. *****Fig. A2.1** Grad-CAM-based prediction without CRF post-processing.**Fig. 5. *****Fig. A2.2** Grad-CAM-based prediction with CRF post-processing.

Observation: Visually, CRF post-processing leads to noticeably sharper and more accurate segmentation boundaries, particularly around fine-grained object contours.

A3. Grad-CAM vs. Bounding Box Supervision

To highlight the differences in boundary quality, we compare segmentation results obtained using Grad-CAM (without CRF) to those derived from bounding boxes.

**Fig. 6. *****Fig. A3.1** Grad-CAM-based prediction without CRF post-processing.**Fig. 7. *****Fig. A3.2** Segmentation using bounding-box-derived masks.

Observation: Bounding box supervision yields tighter object boundaries and cleaner masks, whereas Grad-CAM-only supervision often results in blurry or overly extended edges.